Likelihood-Based Evidential Comparison of Classical Goodness-of-Fit Tests

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Abstract:

The Law of Likelihood establishes that the strength of statistical evidence favoring one hypothesis over another is represented by their likelihood ratio. Within the framework of evidential statistics, this study compares the performance of several classical goodness-of-fit tests—including the Anderson–Darling, Kolmogorov–Smirnov, Cramér–von Mises, Watson, and Likelihood Ratio Test—based on their ability to yield reliable and interpretable evidence. Simulation results across different sample sizes quantify the probabilities of weak and misleading evidence under both null and alternative models, offering new insights into the evidential performance of these tests.

Keywords: Evidential statistics, Goodness-of-fit tests, Law of Likelihood, Misleading evidence, Weak evidence.

Mathematics Subject Classification (2010): 62F25, 62G15.

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1 Introduction

Goodness-of-fit tests are fundamental tools for assessing whether observed data are consistent with a specified probability distribution, and they play an essential role in statistical modeling and inference. Classical goodness-of-fit procedures have been widely studied since the introduction of

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Pearson's chi-square test Pearson (1900), followed by the development of influential methods such as the Kolmogorov–Smirnov Kolmogorov (1933); Smirnov (1948), Cramér–von Mises Cramér (1928); von Mises (1931), Anderson–Darling Anderson and Darling (1952), and Watson statistics. These tests differ in their sensitivity to various types of departures from the hypothesized distribution, including deviations in tails, scale, and overall distributional shape. Despite the widespread use of classical hypothesis testing procedures, traditional statistical methods do not provide a direct and explicit interpretation of statistical evidence in favor of one hypothesis over another Royal (2000). Consequently, evidential statistics has emerged as an alternative framework for quantifying the strength of evidence provided by data. Several studies have investigated measures of statistical support and evidence under different settings Emadi and Arghami (2003); Emadi et al (2007); Doostparast and Emadi (2006); Arashi and Emadi (2008); Habibirad et al (2006); Hashempour (2017); Sarvari and Doostparast (2025). Finally, the structure of the paper is as follows. In Section 2, the concepts of strong and weak statistical evidence are introduced for goodness-of-fit tests within the likelihood-ratio framework. Section 3 presents the simulation study designed to evaluate the evidential performance of the considered goodness-of-fit statistics under different sample sizes and alternative distributions. In Section 4, the proposed evidential approach is illustrated through an analysis of real data, where the strength of statistical evidence provided by the competing models is examined.

2 Evidential goodness-of-fit tests based on likelihood ratios

Let $X_1, \ldots, X_n$ be independent and identically distributed (iid) observations generated from an unknown density f. The likelihood principle states that all information in the data relevant to inference about the model is contained in the likelihood function. For a parametric model indexed by θ, the likelihood can be written as $L(\theta \mid x) = p(x \mid \theta)\, c$, where $p(x \mid \theta)$ is the joint density (or probability mass function) of the observed sample $x = (x_1, \ldots, x_n)$ and $c > 0$ is an arbitrary scaling constant. In continuous settings, probabilities are replaced by probability densities. Throughout, the observed data are treated as fixed, while the model (or its parameters) varies. In the evidential framework, the basic quantity for comparing two competing hypotheses is the likelihood ratio (LR). Consider two hypotheses H_i and H_j. The LR of H_i against H_j is defined by $\mathrm{LR}_{i|j} = \frac{p(x|H_i)}{p(x|H_j)}, \quad i \neq j$. Because the same multiplicative constant c appears in both numerator and denominator, it cancels out, making the LR an intrinsic measure of relative support provided by the data. In goodness-of-fit problems, we take the null hypothesis as $H_0 : F = F_0$, and compare it to an alternative hypothesis H_1 representing departures from F_0. Depending on the application, H_1 may be specified as a simple alternative, a parametric family, or a composite/nonparametric alternative. The evidential analysis then proceeds by evaluating how strongly the data support H_0 relative to H_1 via $\mathrm{LR}_{0|1}$. To operationalize evidential conclusions, it is common to adopt thresholds for strong evidence. Let $k > 1$ denote a user-chosen threshold (depending on the scientific context). Then $\mathrm{LR}_{i|j} \geq k$ is interpreted as strong evidence in favor of H_i over H_j; $\mathrm{LR}_{i|j} \leq 1/k$ is interpreted as strong evidence in favor of H_j over H_i; $1/k < \mathrm{LR}_{i|j} < k$ corresponds to weak or inconclusive evidence. A key feature of the evidential framework is that it distinguishes between (i) obtaining strong evidence for the true hypothesis, (ii) being misled into strong evidence for the false hypothesis, and (iii) obtaining weak evidence. For two hypotheses H_i and H_j and threshold $k > 1$ Under H_j, the probability of observing strong evidence for H_j (i.e., against H_i) is $\mathrm{SEV}_{j|j} = P_{H_j}\big(\mathrm{LR}_{i|j} < \frac{1}{k}\big)$. Under H_j, the probability of observing misleading evidence in favor of H_i is $\mathrm{MEV}_{i|j} = P_{H_j}\big(\mathrm{LR}_{i|j} > k\big)$. Under H_i (and similarly under H_j), the probability of obtaining weak evidence is $W_i = P_{H_i}\big(\frac{1}{k} < \mathrm{LR}_{i|j} < k\big)$. These three regions form a partition of the sample space. Consequently, for a fixed k and $i \neq j$, $P_{H_j}\big(\mathrm{LR}_{i|j} < \frac{1}{k}\big) + P_{H_j}\big(\frac{1}{k} < \mathrm{LR}_{i|j} < k\big) + P_{H_j}\big(\mathrm{LR}_{i|j} > k\big) = 1$. This decomposition provides an interpretable assessment of the operating characteristics of an evidential goodness-of-fit procedure. It clarifies how frequently the method yields decisive evidence, how often it remains inconclusive, and how often it risks strongly favoring an incorrect model.

3 Simulation study

In this Section, the evidential performance of several goodness-of-fit statistics, including Anderson–Darling (AD), Cramér–von Mises (CvM), Kolmogorov–Smirnov (KS), Watson, Shapiro–Wilk (SW), and likelihood-ratio (LR) measures, was investigated using an evidence-based framework. Simulation studies were conducted for sample sizes $n = 10, 20, 30, 50, 70, 100$, and kernel density estimation was employed to approximate the distributions of the statistics under both H_0 and H_1. Based on these estimated densities, LR-based support measures and evidence classifications were obtained for evidence thresholds $k = 8, 16, 32$. The results reported in Table 1 indicate that the AD, CvM, and Watson statistics consistently provide the highest probabilities of correct evidence under both Exponential and Normal null models, while maintaining nearly zero probabilities of misleading evidence. In contrast, the KS statistic shows weak discriminatory power across most scenarios, and the LR statistic demonstrates unstable behavior, particularly under the Normal model. The SW statistic exhibits moderate performance but remains inferior to AD, CvM, and Watson. Figure 1 illustrates the evolution of strong, weak, and misleading evidence probabilities across different sample sizes. As the sample size increases, AD, CvM, and Watson rapidly achieve high levels of strong correct evidence, whereas KS remains dominated by weak evidence. The LR statistic shows model-dependent instability, and SW provides only moderate decisiveness. Furthermore, the proposed upper-bound magnitudes presented in Table 2 display coherent scaling with increasing evidence thresholds. Among the statistics, AD exhibits the strongest proportional increase with k, reflecting its sensitivity to tail deviations, while CvM and Watson show moderate growth and KS remains relatively stable due to its bounded nature. Overall, the findings demonstrate the superiority of AD, CvM, and Watson as reliable and robust statistics for evidence-based goodness-of-fit inference.

4 Real Data Analysis

Table 3 shows times between 48 (in minutes) consecutive telephone calls to a company's switchboard, due to Castillo et al (2005). He showed that the times between the consecutive telephone calls follow the EXP distribution. Based on the comparison with the proposed upper-bound magnitudes and Table 4 the goodness-of-fit statistics provide strong support for the exponential model, with all values falling well within or far below the expected thresholds. Conversely, the normal model exhibits severe violations across all statistics. Therefore, the results clearly indicate excellent agreement with the exponential distribution and substantial evidence against normality.

Table 3: Times(in minutes) between 48 consecutive calls

1.34	0.14	0.33	1.68	1.86	1.31	0.83	0.33
2.20	0.62	3.20	1.38	0.96	0.28	0.44	0.59
0.25	0.51	1.61	1.85	0.47	0.41	1.46	0.09
2.18	0.07	0.02	0.64	0.28	0.68	1.07	3.25
0.59	2.39	0.27	0.34	2.18	0.41	1.08	0.57
0.35	0.69	0.25	0.57	1.90	0.56	0.09	0.28

Table 4: Statistics

H_0	AD	KS	CvM	Watson	LR
Exponential	0.544	0.893	0.085	0.085	0.106
Normal	21.118	0.910	4.172	0.908	68880.976

Discussion and Conclusion

In this paper, we evaluated several classical goodness-of-fit tests for exponential and normal distributions —namely the Anderson–Darling, Cramér–von Mises, Kolmogorov–Smirnov, Watson, and likelihood ratio tests— within a likelihood-based evidential framework. Using kernel density estimation and extensive simulation under various alternative distributions, we quantified the probabilities of strong, weak, and misleading evidence for both the null and the alternative hypotheses. The results show that the Anderson–Darling, Cramér–von Mises, and Watson tests provide the most favorable evidential profiles, combining high probabilities of strong evidence for the true model with very low rates of misleading evidence. In contrast, the Kolmogorov–Smirnov test frequently yields weak or inconclusive evidence, while the likelihood ratio test, although capable of producing strong evidence, is associated with a comparatively higher risk of misleading conclusions.

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Table 1: Mean Estimated Probabilities of Evidence Outcomes for Each Statistic

H_0	Statistic	k	$SEV_{1\|1}$	$SEV_{0\|0}$	$MEV_{1\|0}$	$MEV_{0\|1}$	W_1	W_0
Exponential	AD	8.000	0.888	0.899	0.004	0.006	0.105	0.097
		16.000	0.865	0.861	0.003	0.003	0.132	0.137
		32.000	0.823	0.805	0.001	0.001	0.176	0.194
	CvM	8.000	0.870	0.885	0.005	0.007	0.123	0.110
		16.000	0.844	0.837	0.003	0.003	0.153	0.160
		32.000	0.800	0.784	0.001	0.001	0.199	0.215
	KS	8.000	0.196	0.261	0.007	0.007	0.796	0.732
		16.000	0.148	0.210	0.002	0.003	0.850	0.788
		32.000	0.110	0.167	0.001	0.001	0.889	0.832
	LR	8.000	0.671	0.553	0.098	0.130	0.199	0.349
		16.000	0.630	0.489	0.069	0.111	0.260	0.442
		32.000	0.592	0.430	0.049	0.094	0.314	0.521
	Watson	8.000	0.805	0.787	0.006	0.007	0.188	0.207
		16.000	0.765	0.732	0.003	0.002	0.233	0.265
		32.000	0.733	0.681	0.001	0.001	0.266	0.317
Normal	AD	8.000	0.843	0.837	0.003	0.003	0.154	0.160
		16.000	0.820	0.811	0.001	0.001	0.179	0.188
		32.000	0.800	0.790	0.000	0.000	0.199	0.209
	CvM	8.000	0.761	0.770	0.002	0.001	0.238	0.227
		16.000	0.748	0.761	0.001	0.001	0.251	0.238
		32.000	0.734	0.748	0.001	0.000	0.266	0.252
	KS	8.000	0.285	0.292	0.012	0.008	0.707	0.696
		16.000	0.207	0.216	0.004	0.002	0.790	0.779
		32.000	0.143	0.165	0.002	0.000	0.857	0.833
	LR	8.000	0.699	0.000	0.379	0.000	0.301	0.621
		16.000	0.621	0.000	0.237	0.000	0.379	0.763
		32.000	0.558	0.000	0.149	0.000	0.442	0.851
	ShapiroW	8.000	0.452	0.391	0.007	0.006	0.542	0.603
		16.000	0.406	0.335	0.003	0.002	0.593	0.663
		32.000	0.364	0.297	0.001	0.001	0.635	0.702
	Watson	8.000	0.779	0.778	0.002	0.001	0.220	0.220
		16.000	0.762	0.768	0.001	0.001	0.237	0.231
		32.000	0.749	0.752	0.001	0.000	0.250	0.248

Table 2: Proposed upper-bound magnitudes for the four goodness-of-fit statistics

k	AD	KS	CvM	Watson
8	1	0.90	0.2	0.1
16	2	0.95	0.4	0.2
32	4	0.99	0.8	0.3

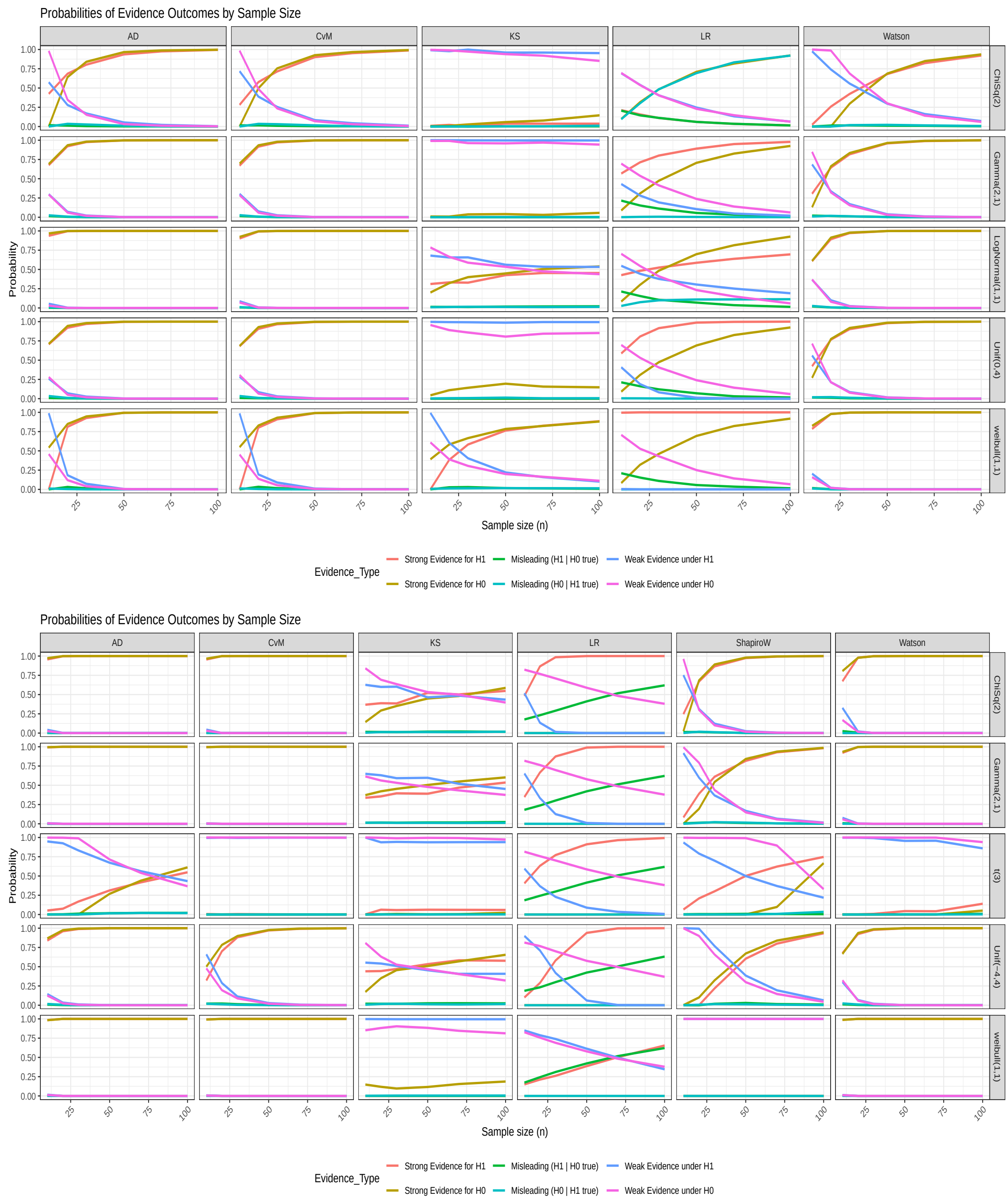

Figure 1: Probabilities of Evidence Outcomes by Sample Size